

The error of the binned Kramers–Moyal estimator

variance, bias, and what measurement noise really does

How $\langle(\hat{b} - b)^2\rangle$ depends on $D, N, \Delta t, h$ and σ_{obs}
 — derived, measured, and checked against a handwritten draft

Companion to `outputs/binned_km/` • every number here is reproduced by
`scripts/09_binned_error_analysis.py`

The whole result in one line. For a cell of width h holding n increments,

$$\underbrace{\frac{2D}{n\Delta t}}_{\text{variance}} + \underbrace{\frac{|\nabla b|^2 h^2}{12}}_{\text{readout bias}} + \underbrace{O(\Delta t^2)}_{\text{discretisation}} + \underbrace{b^2 \left(\frac{\sigma_{\text{obs}}^2}{D\Delta t} \right)^2 + \frac{2\sigma_{\text{obs}}^2 n_{\text{vis}}}{n^2 \Delta t^2}}_{\text{measurement noise}}$$

with $n = M\rho h^d$, so that $n\Delta t$ is the *time* the walkers spend in the cell. Three things in that formula are not in the draft and change the conclusions: n depends on h and on Δt ; the noise term carries $2\sigma^2$, not σ^2 ; and measurement noise acts mainly as a *bias*, not as extra variance.

1 Setup and notation

The data are M recorded transitions $(x_i, \Delta x_i)$ from $dx = b(x) dt + \sqrt{2D} dW$ observed at spacing Δt , so that over one step

$$\Delta x_i = b(x_i) \Delta t + \sqrt{2D\Delta t} \xi_i, \quad \xi_i \sim \mathcal{N}(0, I_d) \text{ i.i.d.} \quad (1)$$

Rung 1 lays a grid of cells of width h , and for the cell C containing x returns

$$\hat{b}(x) = \frac{1}{n\Delta t} \sum_{i \in C} \Delta x_i, \quad n = \#\{i : x_i \in C\}, \quad (2)$$

the same value for every query point in the cell. That last clause is not a detail; it is where the leading bias comes from.

Write $\bar{b}_C$ for the density-weighted average of the true b over C , and $m = \mathbb{E}[\Delta x | C]$ for the true mean increment there. Then

$$\hat{b}(x) - b(x) = \underbrace{\hat{b} - \mathbb{E}\hat{b}}_{\text{(i) statistical}} + \underbrace{\frac{m}{\Delta t} - \bar{b}_C}_{\text{(ii) } \Delta t, \sigma_{\text{obs}}} + \underbrace{\bar{b}_C - b(x)}_{\text{(iii) readout}}. \quad (3)$$

The three are uncorrelated to leading order, so their mean squares add. Everything below is per component; for $|\hat{b} - b|^2$ in d dimensions, multiply the variance by d .

2 The variance

2.1 The algebra, with the one step that is easy to get wrong

Let $S = \sum_{i \in C} \Delta x_i$. Then $\hat{b} = S/(n\Delta t)$ and

$$\mathbb{E}[S^2] = \sum_i \mathbb{E}[\Delta x_i^2] + \sum_{i \neq j} \mathbb{E}[\Delta x_i] \mathbb{E}[\Delta x_j] = n \langle \Delta x^2 \rangle + n(n-1) \langle \Delta x \rangle^2. \quad (4)$$

There are $n(n-1)$ ordered off-diagonal pairs, **not** n^2 : the diagonal has already been counted once, with the *second* moment rather than the square of the first. Everything then collapses:

$$\text{Var}(\hat{b}) = \frac{\mathbb{E}[S^2] - (\mathbb{E}S)^2}{n^2 \Delta t^2} = \frac{n \langle \Delta x^2 \rangle + n(n-1) \langle \Delta x \rangle^2 - n^2 \langle \Delta x \rangle^2}{n^2 \Delta t^2} = \frac{\langle \Delta x^2 \rangle - \langle \Delta x \rangle^2}{n \Delta t^2}. \quad (5)$$

Every $\langle \Delta x \rangle^2$ must cancel. If one survives with a coefficient of order $1/\Delta t^2$, the culprit is almost always n^2 written where $n(n-1)$ belongs: the difference is exactly $-\langle \Delta x \rangle^2/(n \Delta t^2) = -b^2/n$, and losing it leaves a spurious $+b^2/\Delta t^2$ -sized term behind.

From (1), with b treated as constant across the cell, $\langle \Delta x^2 \rangle - \langle \Delta x \rangle^2 = 2D\Delta t + \Delta t^2 \text{Var}_C(b)$, and the second piece is smaller than the first by $\Delta t |\nabla b|^2 h^2/(24D) \sim 10^{-6}$ at every setting used in this project. Hence

$$\boxed{\text{Var}(\hat{b}) = \frac{2D}{n \Delta t} \quad \text{per component.}} \quad (6)$$

A cell averages noise of size $\sqrt{2D/\Delta t}$ down by $\sqrt{n}$. Nothing else enters: not h , not ρ , not b — *explicitly*.

2.2 n is not a free parameter: where h and Δt actually enter

The count is set by the experiment,

$$n \simeq M \rho(x) h^d, \quad M = N_{\text{walkers}} \times T_{\text{steps}}, \quad (7)$$

so (6) becomes the statement that matters:

$$\boxed{\text{Var}(\hat{b}) = \frac{2D}{M \rho h^d \Delta t} = \frac{2D}{T_{\text{occ}}}, \quad T_{\text{occ}} = n \Delta t = \text{time spent in the cell.}} \quad (8)$$

Δt cancels. Sampling twice as fast makes every $\Delta x/\Delta t$ a factor $\sqrt{2}$ noisier and hands you twice as many of them. At fixed total observation time the variance is unchanged. The sampling rate is therefore *not* a variance knob; it is a bias knob (Section 3) and, once positions are measured imperfectly, a catastrophe knob (Section 5).

h enters only through the count, and it enters hard. $\text{Var} \propto h^{-d}$: halving the cell width doubles the variance in 1D, quadruples it in 2D, and multiplies it by 2^d in general. This is the entire variance side of the bias–variance tradeoff, and it is invisible if n is carried around as an independent symbol.

3 The bias

3.1 Readout bias: first order in h , and particular to this estimator

The cell’s own average $\bar{b}_C$ is accurate to $O(h^2)$ *at the cell centre* — the linear term cancels by symmetry. But `predict` returns that one number for every query in the cell, so at a query displaced by δ from the centre,

$$\bar{b}_C - b(x) = -J_b \delta + O(h^2), \quad \langle |J_b \delta|^2 \rangle = \frac{|\nabla b|^2 h^2}{12} \quad \text{per component}, \quad (9)$$

since a uniform δ over $[-h/2, h/2]$ has variance $h^2/12$. The bias is **first order in h** , and this is not a general property of local averaging: a kernel centred on the query is $O(h^2)$ everywhere. Measured on the 2D random field with noise-free targets, the error against cell width has log-log slope 0.97 at random query points and 1.84 at cell centres. That gap is most of why rung 2 beats rung 1.

3.2 Discretisation bias: first order in Δt

Independently of the binning, the target itself is only asymptotically b :

$$\frac{1}{\Delta t} \mathbb{E}[\Delta x | x] = b(x) + \frac{\Delta t}{2} [(b \cdot \nabla) b + D \nabla^2 b] + O(\Delta t^2). \quad (10)$$

This is the one term that punishes large Δt , and it is why the simulator integrates with substeps sub-steps while recording at Δt : otherwise an $O(\Delta t)$ error in the estimator is indistinguishable from an $O(\Delta t)$ error in the integrator.

4 The total, the optimal cell width, and the curse of dimensionality

Adding (8) and (9), and averaging over query points drawn from ρ (so $\mathbb{E}_\rho[1/\rho] = |\Omega|$, the box volume):

$$\boxed{\text{MSE}(h) \simeq \underbrace{\frac{2D|\Omega|}{M \Delta t h^d}}_{A/h^d} + \underbrace{\frac{\langle |\nabla b|^2 \rangle_\rho}{12}}_{B h^2} h^2 + O(\Delta t^2)}. \quad (11)$$

Minimising,

$$h^* = \left(\frac{dA}{2B} \right)^{\frac{1}{d+2}} \propto \left(\frac{D}{M \rho \Delta t |\nabla b|^2} \right)^{\frac{1}{d+2}}, \quad \text{bins per axis} \propto M^{\frac{1}{d+2}}, \quad \text{MSE}^* \propto M^{-\frac{2}{d+2}}. \quad (12)$$

The exponent $1/(d+2)$ is exactly the `auto_bins` rule in `dfi/estimators/local.py`. It is $1/(d+2)$ and not the textbook $1/(d+4)$ *because* the readout bias is first order: an $O(h^2)$ bias would give $M^{1/(d+4)}$, which is the right rule for a kernel and the wrong one for a grid of constants. And $\text{MSE}^* \propto M^{-2/(d+2)}$ is the curse of dimensionality in one line: to halve the error in d dimensions you need $2^{(d+2)/2}$ times the data — $4\times$ at $d=2$, $16\times$ at $d=6$.

5 Measurement noise

Now suppose positions are observed as $\tilde{x}_k = x_k + \varepsilon_k$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ i.i.d. The observed increment is

$$\widetilde{\Delta x}_k = \Delta x_k + \varepsilon_{k+1} - \varepsilon_k. \quad (13)$$

The noise enters in two places — the increment *and* the cell assignment — and the second one is far more destructive.

5.1 (a) Extra variance: $2\sigma^2$, reduced by telescoping

A *difference* of two independent draws has variance $2\sigma^2$, not σ^2 . Naively,

$$\text{Var}(\hat{b}) = \frac{2D}{n\Delta t} + \frac{2\sigma^2}{n\Delta t^2} = \frac{2D}{T_{\text{occ}}} + \frac{2\sigma^2}{T_{\text{occ}} \Delta t}. \quad (14)$$

Note the asymmetry: the diffusion term is blind to Δt , the noise term diverges as $\Delta t \rightarrow 0$. Measure faster and you buy nothing but measurement noise.

But consecutive increments *inside the same cell* share their ε , and the sum telescopes: over a contiguous visit, $\sum(\varepsilon_{j+1} - \varepsilon_j) = \varepsilon_{\text{out}} - \varepsilon_{\text{in}}$. Only the two endpoints survive, so

$$\text{Var}(\hat{b}) = \frac{2D}{n\Delta t} + \frac{2\sigma^2 n_{\text{vis}}}{n^2 \Delta t^2}, \quad \frac{n}{n_{\text{vis}}} \approx \frac{h^2}{2dD\Delta t} \text{ steps per visit}. \quad (15)$$

5.2 (b) The bias that actually kills the estimator

The cell is chosen with $\tilde{x}$, so the estimator *selects on* ε_k — and $-\varepsilon_k$ sits inside the increment it then averages. If the density is locally flat the selection is symmetric and nothing happens; with a density gradient,

$$\mathbb{E}[\varepsilon_k \mid \tilde{x}_k = x] \simeq \sigma^2 \nabla \log \rho(x), \quad (16)$$

so that, using $-\varepsilon_k/\Delta t$ and the equilibrium identity $\nabla \log \rho = b/D$,

$$\mathbb{E} \hat{b} \simeq b + \frac{\sigma^2}{\Delta t} \nabla \log \rho = b \left(1 + \frac{\sigma^2}{D \Delta t} \right). \quad (17)$$

This is a multiplicative inflation of the drift, and no amount of data removes it. At the project's settings ($D = 0.3$, $\Delta t \approx 10^{-3}$) a position error of $\sigma = 0.02$ — one percent of the box — gives $\sigma^2/(D\Delta t) = 1.33$, i.e. an estimate $2.33\times$ too large, i.e. $\text{nrmse} \approx 1.33$ on its own. Every trajectory method in the ladder scores above 1.2 at that noise level, and this is why. It also explains the shape of the dependence: the damage scales as $\sigma^2/\Delta t$, so it is the *fast-sampling* experiments that suffer most.

6 Numerical verification

Every claim above is measured on a 1D periodic field $b(x) = -\sin \pi x$ on $[-1, 1]$ with $D = 0.3$, $\Delta t = 10^{-3}$, where $\rho_{\text{ss}} \propto \exp(\frac{1}{\pi D} \cos \pi x)$ is known in closed form. $R = 64$ *independent replicas* of 200×4000 steps are simulated side by side, so the variance of a per-cell estimate is measured across replicas rather than inferred from a single fit. Run `python scripts/09_binned_error_analysis.py`.

6.1 Variance and the U curve

bins	h	n per cell	Var measured	$2D/(n\Delta t)$	readout ² measured	$\langle \nabla b ^2 \rangle h^2/12$
4	0.500	199 947	0.00285	0.00300	0.0920	0.1150
8	0.250	85 159	0.00639	0.00600	0.0274	0.0288
16	0.125	39 708	0.01209	0.01200	0.00709	0.00719
32	0.0625	19 436	0.02411	0.02400	0.00179	0.00180
64	0.0313	9 665	0.04756	0.04800	0.00045	0.00045
128	0.0156	4 825	0.09557	0.09600	0.00011	0.00011

The per-cell ratio of measured to predicted variance has median 0.97–1.09 across all six widths (0.99 pooled over all 252 cells). The variance doubles at every halving of h , as h^{-d} with $d = 1$ demands. The readout bias follows Bh^2 with $B = \langle |\nabla b|^2 \rangle_\rho/12 = 0.460$ to within 1% for $h \leq 0.125$, and falls short at $h = 0.5$ where the quadratic approximation to a sine gives up.

With $M = 8 \times 10^5$ this predicts $h^* = 0.118$, i.e. **17.0 bins**; the measured total error is minimised at **16 bins** (the nearest available). The optimum is shallow on the fine-grid side and steep on the coarse side, which is the usual shape: over-smoothing is worse than under-smoothing for a piecewise-constant fit.

6.2 Measurement noise

σ_{obs}	$1 + \sigma^2/(D\Delta t)$	measured $\hat{b}/\bar{b}$	steps per visit	$\frac{\text{Var}}{\text{diffusion only}}$	$\frac{\text{Var}}{\text{naive } 2\sigma^2/n\Delta t^2}$	$\frac{\text{Var}}{\text{telescoped}}$
0	1.000	1.001	6.3	0.97	0.97	0.97
0.005	1.083	1.087	6.1	1.15	1.06	1.14
0.010	1.333	1.320	5.5	1.81	1.36	1.71
0.020	2.333	2.350	4.1	4.18	1.79	3.17
0.020, cell chosen with the <i>true</i> x		0.993	6.3	1.25	0.54	1.03

The inflation formula (17) is confirmed to better than 1.5% over a factor of 5 in σ . The last row is the decisive control: feed the estimator the same noisy increments but choose the cell with the true position, and

- the bias **vanishes** (0.993 instead of 2.350) — so the damage is done by the *binning*, not by the noise in the increment;
- the variance matches the telescoped formula (15) to 3%, while the naive $2\sigma^2/(n\Delta t^2)$ overestimates it by a factor 1.9 — telescoping is real;
- with noisy binning the variance is instead $4.2\times$ the diffusion term, and panel (d) shows it stays between $3\times$ and $4.4\times$ at every cell width from $h = 0.016$ to $h = 0.5$. Neither formula accounts for it. The mechanism is the same selection that produces the bias — near a cell edge, membership and the $-\varepsilon_k/\Delta t$ carried by the increment are correlated — but I have no closed form for its size, and the measured width-independence is not what a naive edge-fraction argument ($\propto \sigma/h$) would give. It is the one term in this note that is measured and not derived.

6.3 Against the project’s own sweeps

An independent check on the 2D random field, from `outputs/local_study/data/sweep_obs_noise.csv` (256 walkers, 4000 steps, $D = 0.3$, three seeds with slightly different stability-chosen Δt). Predicted nrmse is $\sigma^2/(D\Delta t)$ from (17), combined in quadrature with the noise-free error of the same run:

Δt	$\sigma^2/(D\Delta t)$ at $\sigma = 0.02$	noise-free nrmse	predicted	measured
1.051×10^{-3}	1.268	0.446	1.344	1.486
0.787×10^{-3}	1.695	0.514	1.771	1.933
1.211×10^{-3}	1.101	0.458	1.192	1.267

Within 7–10% across a $1.5\times$ range of Δt , in a different dimension and a different field family, with no fitted constant. The residual is the edge-selection variance of the previous table, which the prediction omits.

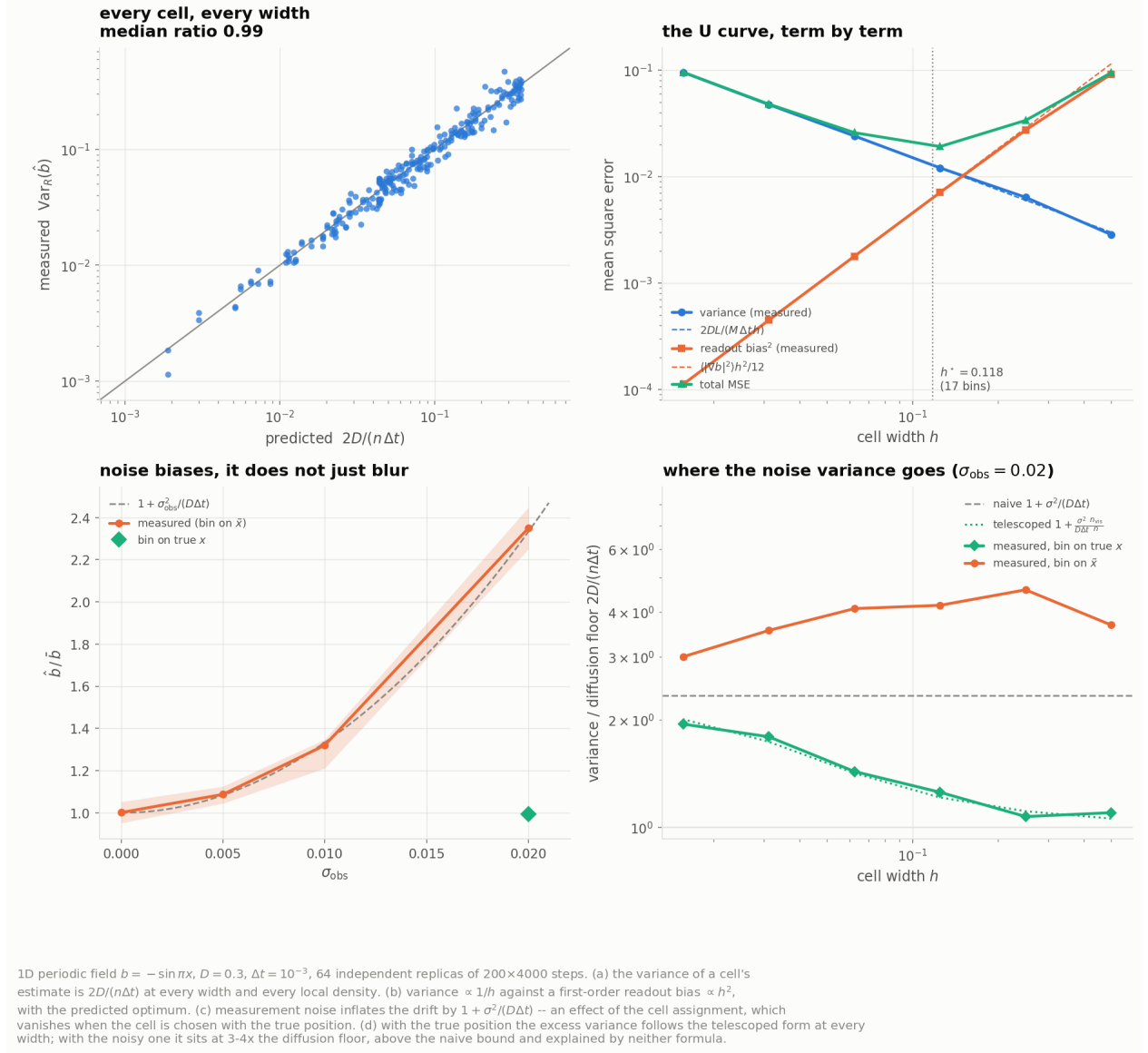


Figure 1: (a) the variance of a cell's estimate equals $2D/(n\Delta t)$ at every cell width and every local density — every point is one cell of one grid, 252 in all. (b) the two terms of (11) measured separately against their predictions, and their sum, with the predicted optimum marked. (c) measurement noise inflates the drift by $1 + \sigma^2/(D\Delta t)$; the green diamond is the control in which the increment is noisy but the cell is chosen with the true position. (d) the same control across cell widths: with the true position the excess variance follows the telescoped form (15); with the noisy one it sits at 3-4x the diffusion floor at every width, above the naive bound.

7 Notes on the handwritten sheet

Figure 2: The draft under discussion. Left column: the variance of the cell mean. Right column: the bin-width bias.

What is right. The structure is the right structure: variance of a sample mean on the left, a Taylor expansion of b across the cell on the right. The conditional-mean definition $b(x) = \frac{1}{\Delta t} \mathbb{E}[\Delta x | x]$ and its binned version $b_h(x) = \frac{1}{\Delta t} \mathbb{E}[\Delta x | x \in [x_0 - \frac{h}{2}, x_0 + \frac{h}{2}]]$ are exactly right, and so is expanding with the Jacobian, $b(x_0 + h) = b(x_0) + J_b \vec{h}$, rather than a scalar derivative. Reaching a $\frac{1}{N\Delta t^2} \langle \Delta x^2 \rangle$ -shaped answer with a $2D$ term and a σ^2 term in it means the skeleton is sound.

The last two lines of the right-hand column, $b(x) - \hat{b} = b(x) - \frac{1}{N\Delta t} \sum_i \Delta x_i (x+h)$, are the start of exactly the right move — putting the bias and the estimator into one expression instead of two separate columns. Carried through, that is (11), and it is where the h -dependence of N becomes unavoidable. It is worth finishing.

Six corrections

1. $(\sum \Delta x)^2$: **the cross terms are $N(N-1)$, not $2N$ (nor N^2).** The sheet has $\frac{1}{\Delta t^2 N^2} (N \langle \Delta x^2 \rangle + 2N \langle \Delta x \rangle^2)$; it should be $\frac{1}{\Delta t^2 N^2} (N \langle \Delta x^2 \rangle + N(N-1) \langle \Delta x \rangle^2)$, as in (4). The consequence is visible two lines later: the $-\frac{1}{\Delta t^2} \langle \Delta x \rangle^2$ from the first line never cancels, and the final expression still carries $(\frac{2}{N\Delta t^2} - \frac{1}{\Delta t^2}) \langle \Delta x \rangle^2$. That is a term of size b^2 , i.e. the *entire drift squared* — not a small correction, and negative. With the right count everything collapses to $\text{Var}(\Delta x)/(N\Delta t^2)$ and no $\langle \Delta x \rangle^2$ survives.
2. $\langle \Delta x^2 \rangle = \langle \Delta x \rangle^2 + 2D + \sigma^2$ **is missing a Δt .** It is $2D\Delta t$: the mean square displacement of a Brownian step grows *linearly in time*. Dimensionally, $[\Delta x^2] = \text{length}^2$ while $[D] = \text{length}^2/\text{time}$. Carried through, the sheet's diffusion term becomes $\frac{2D}{N\Delta t^2}$ where it should be $\frac{2D}{N\Delta t}$ — a factor Δt , which at $\Delta t = 10^{-3}$ is a factor of a thousand in the variance, and it is the term that decides everything.
3. **The measurement-noise term is $2\sigma^2$, not σ^2 .** The observed increment is $\Delta x + \varepsilon_{k+1} - \varepsilon_k$, a *difference* of two independent draws (13). And even $2\sigma^2$ per increment is too many: consecutive steps inside a cell telescope, leaving $2\sigma^2$ per *visit* (15), which at these settings is a further factor of ≈ 6 . Measured: the naive form overestimates by $1.9\times$, the telescoped form is exact to 3%.

4. **h never appears in the variance, because N is carried as a free symbol.** This is the structural gap, not an algebra slip. $N = M\rho h^d$ is the *only* way h reaches the variance, and without it there is no tradeoff against the bias on the right-hand column, no optimal cell width, and no way to answer “how many bins?”. The two columns of the sheet are the two halves of (11) and they need to be added, in the same variables.
5. **Δt cancels in the diffusion term and does not cancel in the noise term.** Substituting $N = M\rho h^d$ turns $\frac{2D}{N\Delta t}$ into $\frac{2D}{M\rho h^d \Delta t}$, which at fixed *total time* $M\Delta t$ is independent of the sampling rate, while $\frac{2\sigma^2}{N\Delta t^2}$ still carries one power of $1/\Delta t$. The sheet treats both as $\frac{1}{N\Delta t^2}$ -type terms with N independent of Δt , which hides the single asymmetry that decides how fast one should sample.
6. **σ_{obs} appears only in the variance, and its main effect is a bias.** At $\sigma = 0.02$ the inflation (17) contributes 1.33 to the nrmse on its own, against 0.4–0.8 from the extra variance (the quadrature residuals in §6.3). Both matter; only one of the two is in the sheet, and it is the smaller. The mechanism is not in the sheet’s model at all, because the sheet conditions on the *true* cell membership: the damage comes from the cell being chosen with $\tilde{x}$. That is the difference between the last two rows of the table on the previous page — 0.993 versus 2.350 — and it is the whole story of why the snapshot methods win at $\sigma_{\text{obs}} = 0.02$ while every trajectory method in the ladder collapses.

Two things simply not there

- The $O(\Delta t)$ discretisation bias $\frac{\Delta t}{2}[(b \cdot \nabla)b + D\nabla^2 b]$, which is independent of N , h and σ and sets the floor at large Δt .
- The factor d : in d dimensions $\mathbb{E}|\hat{b} - b|^2 = \frac{2dD}{N\Delta t} + \dots$, since each component carries its own noise.

One refinement on the right-hand column, which is otherwise correct

$b_h - b \simeq J_b \vec{h}$ is the right leading term *for this estimator*, and it is worth being precise about why. Averaged over the cell, $\bar{b}_C$ is accurate to $O(h^2)$: the linear term cancels because the cell is symmetric about x_0 . The first order survives only because the estimator is asked for b at a query point $x = x_0 + \delta$ and answers with the cell’s single value. So the right statement is (9): the displacement is δ , uniform over the cell, giving $|\nabla b|^2 h^2 / 12$ rather than $|\nabla b|^2 h^2$. The factor $1/12$ matters: with it h^* is 17 bins, which is what the measurement shows; without it, 39.

This distinction is exactly the gap that rung 2 collects. A kernel centred on the query point has no displacement to pay for, so its bias is $O(h^2)$ everywhere: log-log slope 1.85 measured, against 0.97 for the grid. It is also why the two estimators want different bin-count rules, $M^{1/(d+2)}$ against $M^{1/(d+4)}$.

8 The whole budget, assembled

term	size	what changes it
statistical variance	$\frac{2dD}{M\rho h^d\Delta t} = \frac{2dD}{T_{\text{occ}}}$	more walkers, longer runs, wider cells, smaller D . <i>Not</i> the sampling rate.
readout bias	$\frac{ \nabla b ^2 h^2}{12}$	narrower cells; or a kernel, which has $O(h^2)$ bias, i.e. $O(h^4)$ here
discretisation bias	$\frac{\Delta t}{2} [(b \cdot \nabla)b + D\nabla^2 b]$	sampling faster — the only term that wants small Δt
noise variance	$\frac{2\sigma^2 n_{\text{vis}}}{n^2 \Delta t^2}$	sampling <i>slower</i> ; wider cells (more telescoping)
noise bias	$b \frac{\sigma^2}{D\Delta t}$	sampling slower; better metrology. No amount of data helps

Reading the table as an experimental design: Δt is pulled down by the discretisation bias and pushed up, far harder, by everything involving σ_{obs} ; h is pulled down by the readout bias and up by the variance, balancing at $M^{1/(d+2)}$ bins per axis; and M buys error only as $M^{-1/(d+2)}$, which is why the ladder does not stop at rung 1.